NEWTON'S SECOND LAW

02/03/2004

$F = ma = \frac{dp}{dt}$ derived based on observation and experimentation

$ma = 0$ (no external force)

$m\ddot{x} = 0$

$\dot{x} = v = dx/dt$

$\ddot{x} = a = d^2x/dt^2$

$x = at + b$ integration constants

$\dot{x} = a$

$\ddot{x} = 0$

LIMITATION

(i) Non inertial frame. Example - rotating frame (Pseudo forces)

(ii) Zero mass particles (photons)

(iii) Speed of particle tends to speed of light.

CONSERVATION LAWS

Conservation of total energy $E = \int_1^2 F\,dx$

$\rightarrow KE = \frac{1}{2}mv^2$

$\rightarrow U = PE$

$F = -\frac{dU}{dx}$

$E = KE + U$

$= \frac{1}{2}mv^2 + U(x)$

$\frac{dE}{dt} = 0$

$U = U(x, t, v)$ possible

$KE = (x, t, v)$ possible

Assume total energy constant (conserved)

$E = \frac{1}{2}m\dot{x}^2 + U(x)$

$= \frac{1}{2}\left(\frac{dx}{dt}\right)^2 + U(x)$

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$$\frac{dx}{dt} = \pm\sqrt{\frac{2}{m}\left(E - U(x)\right)}$$

$$\int dt = \int \frac{\pm\, dx}{\sqrt{\frac{2}{m}\left(E - U(x)\right)}}$$

$$dt = \frac{\pm\, dx}{\sqrt{\frac{2}{m}\left(E - U(x)\right)}}$$

(i) $U(x) = 0$ $\quad x(t) = at + b$

(ii) $U(x) = \frac{1}{2}kx^2$ $\quad x(t) = A\sin(\omega t + \phi)$

(iii) $E = U(x)$

$$\int dt = \pm\int \frac{dx}{\sqrt{\frac{2}{m}\left(E - U(x)\right)}}$$

Taylor series expansion

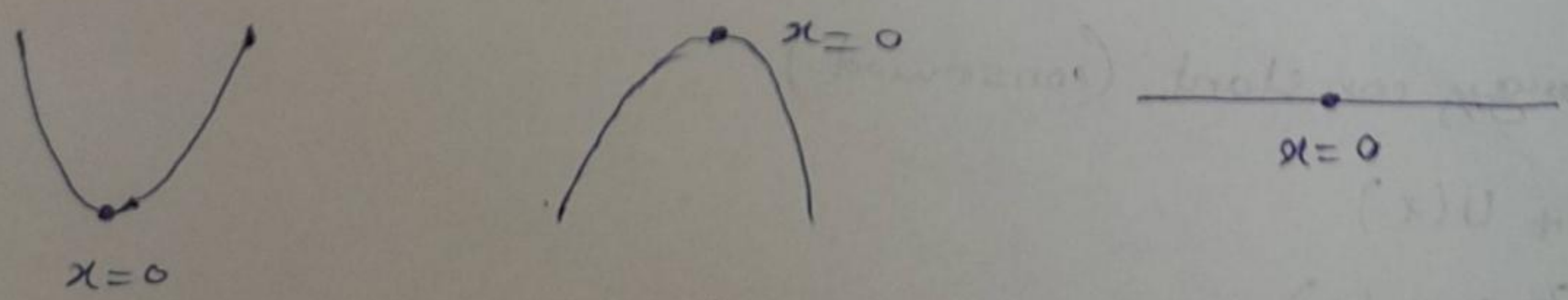
$$U(x) = U(x)\Big|_{x=0} + x\left.\frac{dU}{dx}\right|_{x=0} + \frac{x^2}{2!}\left.\frac{d^2U}{dx^2}\right|_{x=0} + \frac{x^3}{3!}\left.\frac{d^3U}{dx^3}\right|_{x=0} + \cdots$$

If $x=0$ is an equilibrium point

then $\left.\frac{dU}{dx}\right|_{x=0} = 0$

three possibilities

$\frac{d^2U}{dx^2} > 0$ $\qquad$ $\frac{d^2U}{dx^2} < 0$ $\qquad$ $\frac{d^2U}{dx^2} = 0$



stable equilibrium $\qquad$ unstable equilibrium $\qquad$ neutral

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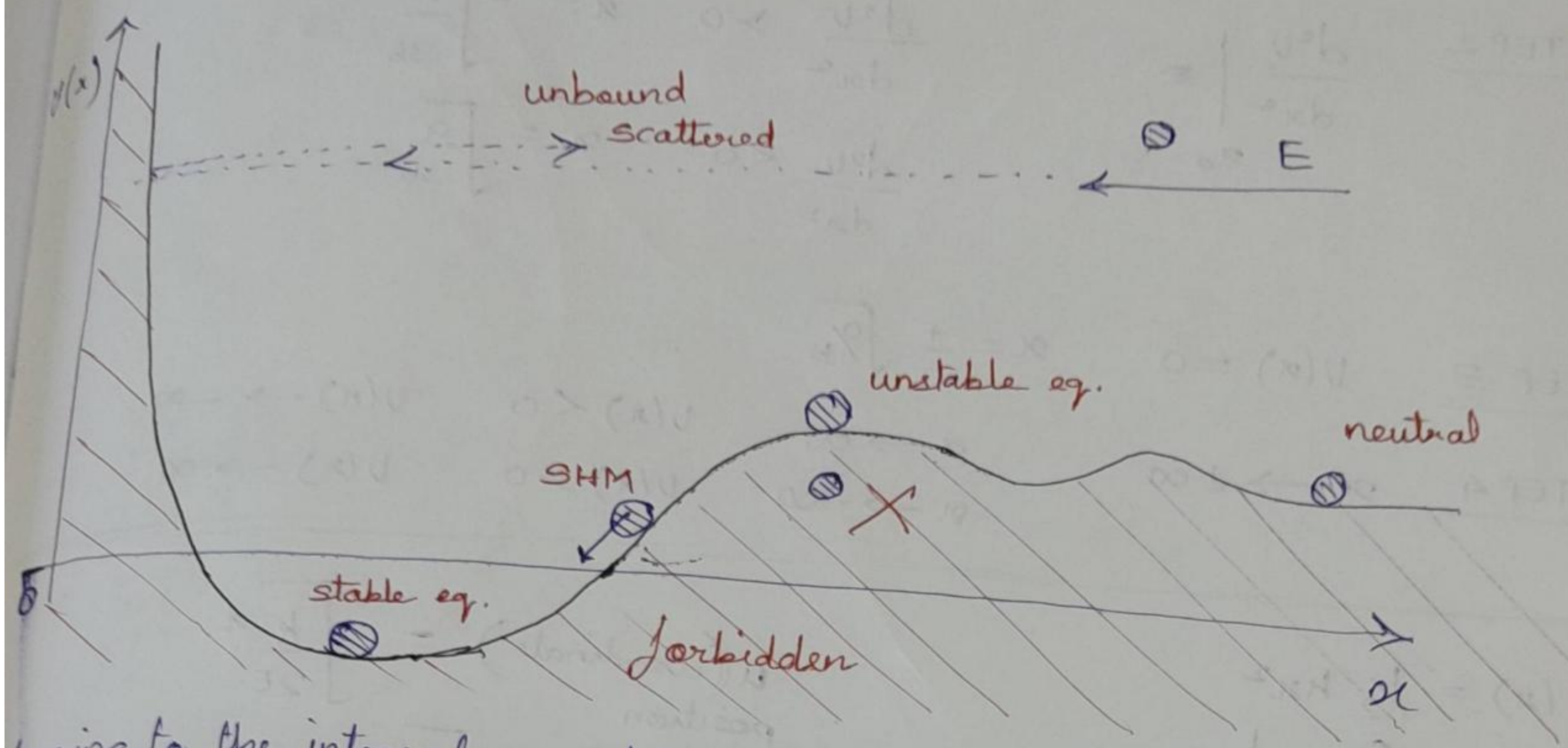
For equilibrium points $dU/dx = 0$

Check for stable/unstable equilibrium



Referring to the integral, particle can never be below the $U(x)$ vs. x graph as $E > U(x)$

i) $U(x) = ax - bx^3$

$\frac{dU(x)}{dx} = a - 3bx^2$

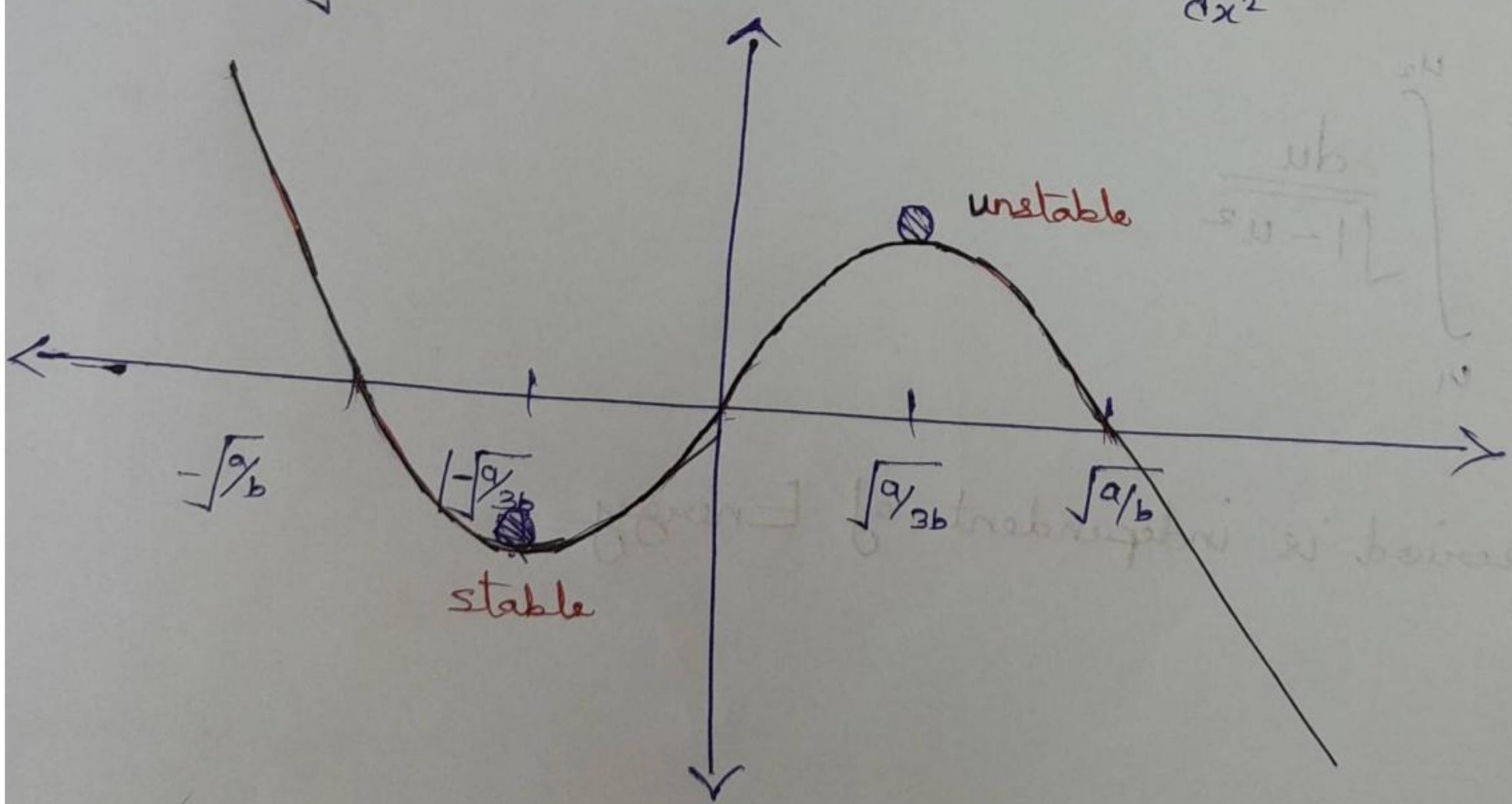
$x = \pm\sqrt{\frac{a}{3b}}$

$x(a - bx^2)$

$x = 0$

$x = \pm\sqrt{\frac{a}{b}}$

$\frac{d^2U}{dx^2} = -6bx$



$> 0 \quad x < 0$

$\frac{d^2U}{dx^2} < 0 \quad x > 0$

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STEP 1 $\frac{dU}{dx} = 0$ $\quad x = \pm\sqrt{\frac{a}{3b}}$

STEP 2 $\left.\frac{d^2U}{dx^2}\right|_{x_0} =$ $\quad \frac{d^2U}{dx^2} > 0 \quad x = -\sqrt{\frac{a}{3b}}$

$\frac{d^2U}{dx^2} < 0 \quad x = \sqrt{\frac{a}{3b}}$

STEP 3 $U(x) = 0$ $\quad x = \pm\sqrt{a/b}$

STEP 4 $x \to \pm\infty$ $\quad x \to \infty \quad U(x) < 0 \quad U(x) \to -\infty$

$x \to -\infty \quad U(x) > 0 \quad U(x) \to \infty$

$U(x) = \frac{1}{2} k x^2$

u (coordinate position) $= \sqrt{\frac{k}{2E}}\, x$

$du = \sqrt{\frac{k}{2E}}$

$$t = \pm \int \frac{dx}{\sqrt{\frac{2E}{m} - \frac{\cancel{2}}{m}\frac{kx^2}{\cancel{2}}}}$$

$$= \int \sqrt{\frac{m}{\cancel{2E}}} \frac{du}{\sqrt{1-u^2}} \sqrt{\frac{\cancel{2E}}{k}}$$

$$t = \pm \int \frac{dx}{\sqrt{\frac{2E}{m}\left(\sqrt{1 - \frac{k}{\cancel{m}}\frac{\cancel{m}}{2E}x^2}\right)}}$$

$$t = \sqrt{\frac{m}{k}} \int_{u_1}^{u_2} \frac{du}{\sqrt{1-u^2}}$$

Time period is independent of Energy

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$U(x) = -k/x$

$$t = \pm\int \frac{dx}{\sqrt{\frac{2E}{m} + \frac{2}{m}\frac{k}{x}}} = \pm\int \frac{dx}{\sqrt{\frac{2E}{m}\left(\sqrt{1 + \frac{\cancel{2}}{\cancel{m}}\frac{k}{x}\frac{\cancel{m}}{\cancel{2}E}}\right)}}$$

$$= \frac{dx}{\sqrt{\frac{2E}{m}\left(\sqrt{1 + \frac{k}{xE}}\right)}}$$

$$u = \frac{xE}{k}$$

$$du = \frac{E}{k}\,dx$$

$$= \sqrt{\frac{m}{2E}}\,\frac{k}{E}\int \frac{du}{\sqrt{1 + \frac{1}{u}}}$$

$$= \sqrt{\frac{mE}{2}}\,\frac{1}{k}\int \frac{\sqrt{u}\,du}{\sqrt{1+u}}$$

extra H

$$t \propto E^{-3/2}$$

09/03/2023

$E = \frac{1}{2}mv^2 + U(x)$

we are assuming energy is conserved.

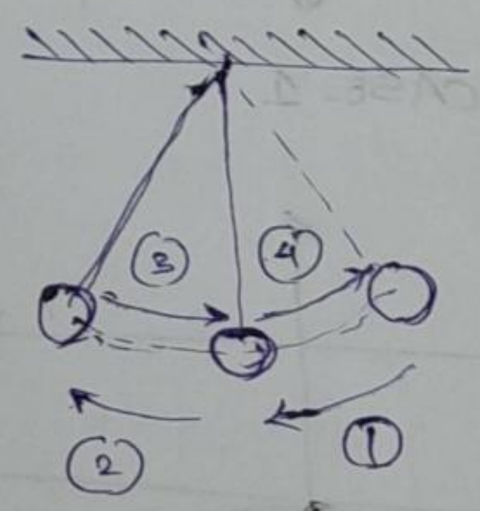
$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$

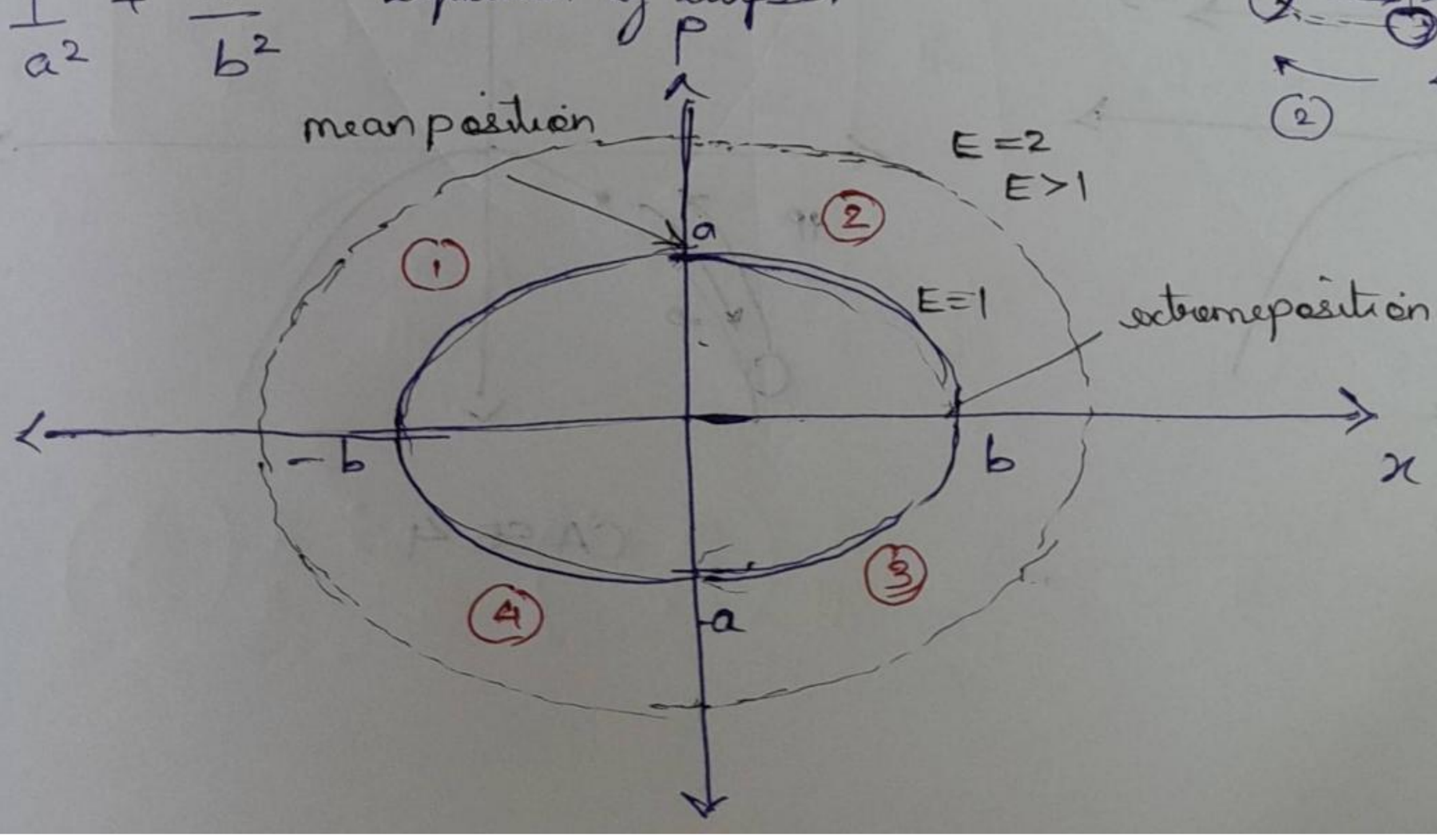
$$E = \frac{p^2}{2m} + \frac{kx^2}{2}$$

$$= \frac{p^2}{2m} + \frac{1}{2}kx^2$$

$a = \sqrt{2m}$ $b = \sqrt{2/k}$

$$E = \frac{p^2}{a^2} + \frac{x^2}{b^2}$$ equation of ellipse.





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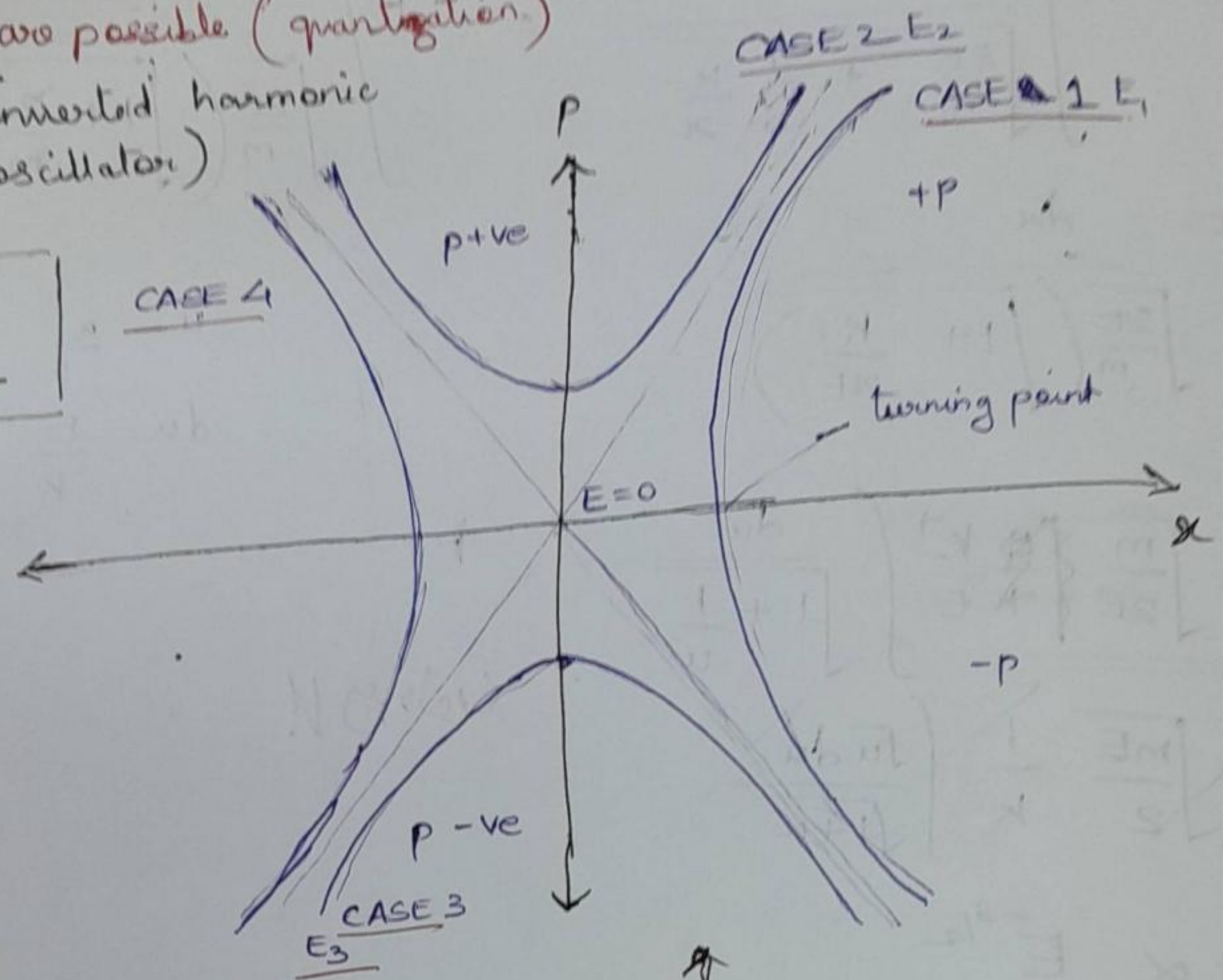
In case of frictional force and damping, phase diagram will look like spiral.
There can be ∞ no. of ellipses infinitesimally close to one another.
Not every ellipse are possible (quantization)

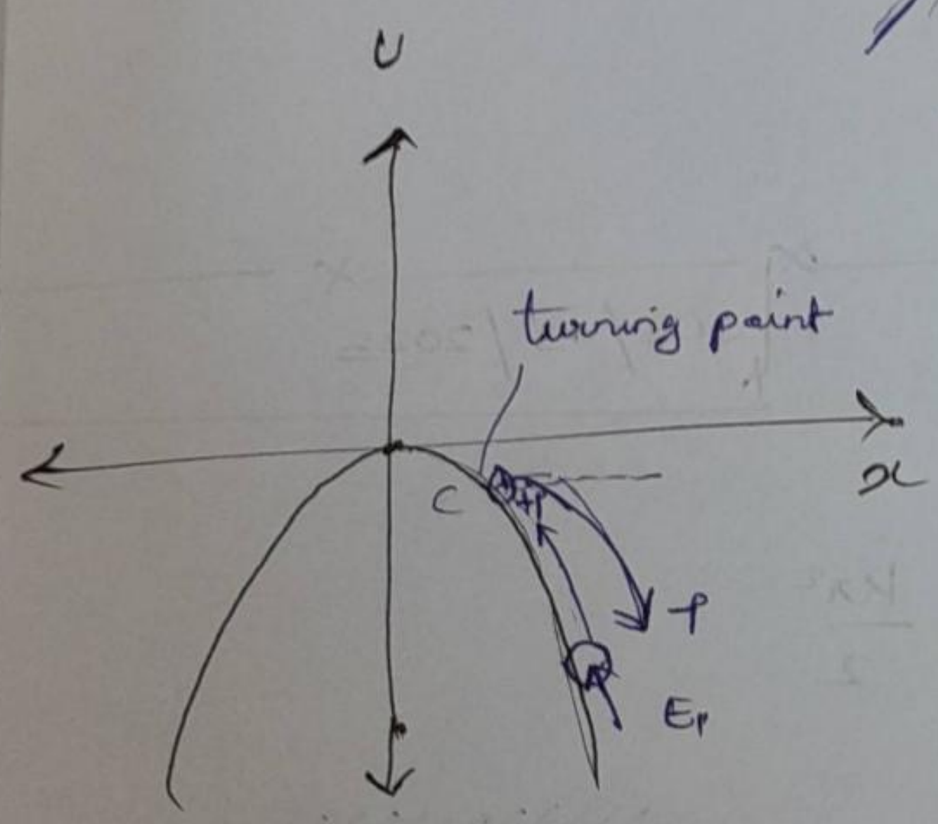
$U(x) = -\frac{1}{2}kx^2$ (inverted harmonic oscillator)

$$\boxed{E = \frac{p^2}{a^2} - \frac{x^2}{b^2}}$$

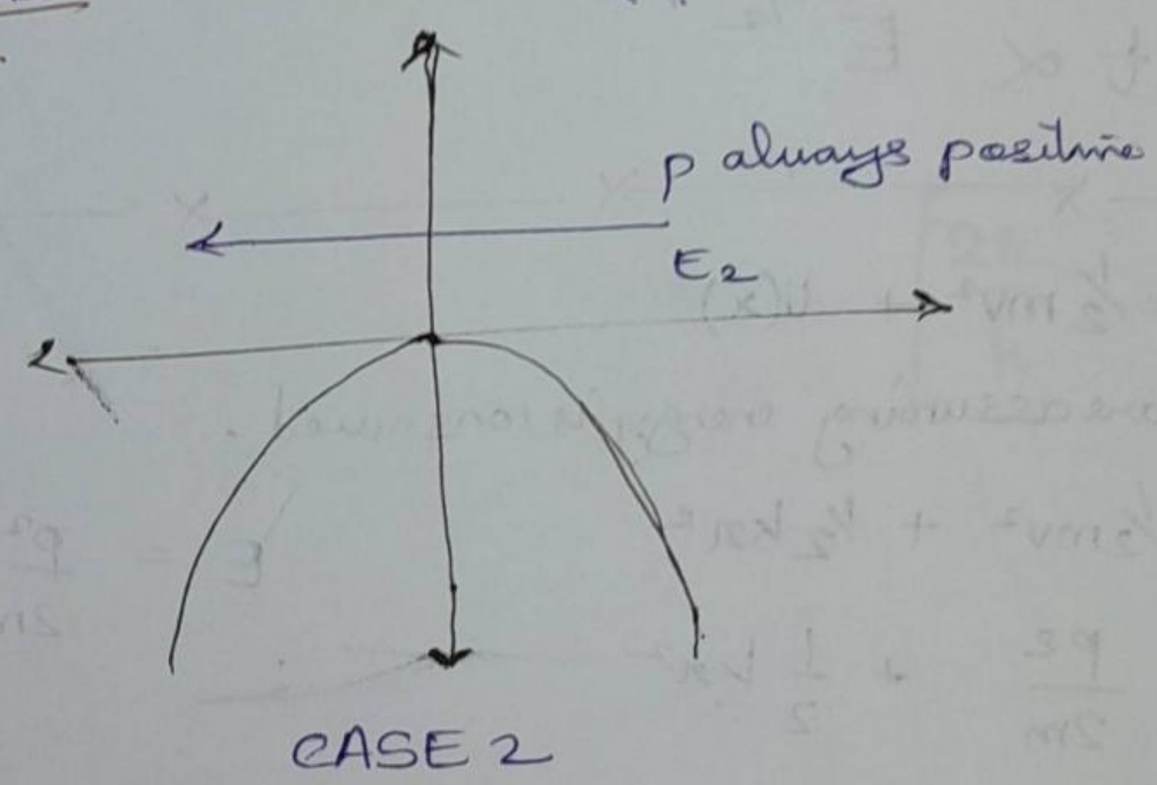
$a = \sqrt{2m}$

$b = \sqrt{2/k}$

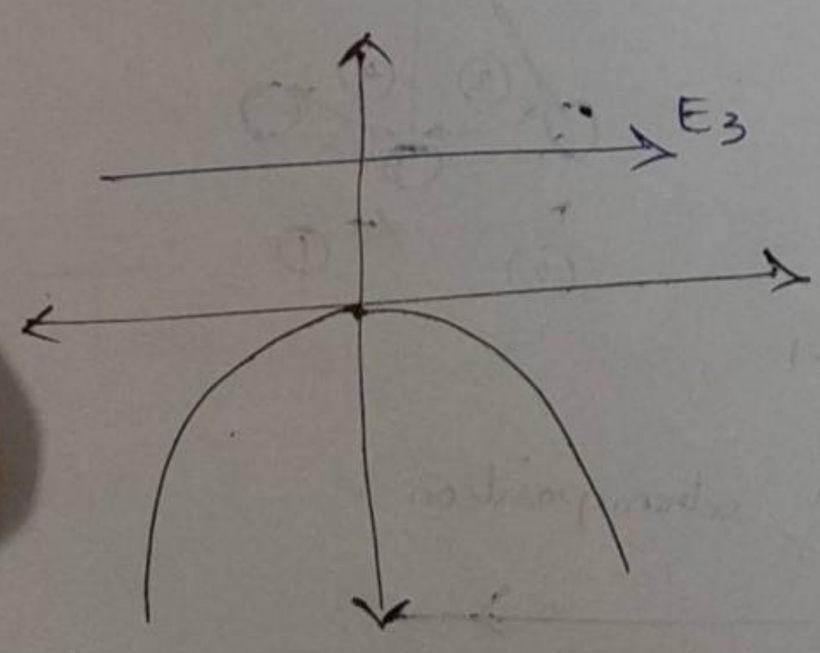




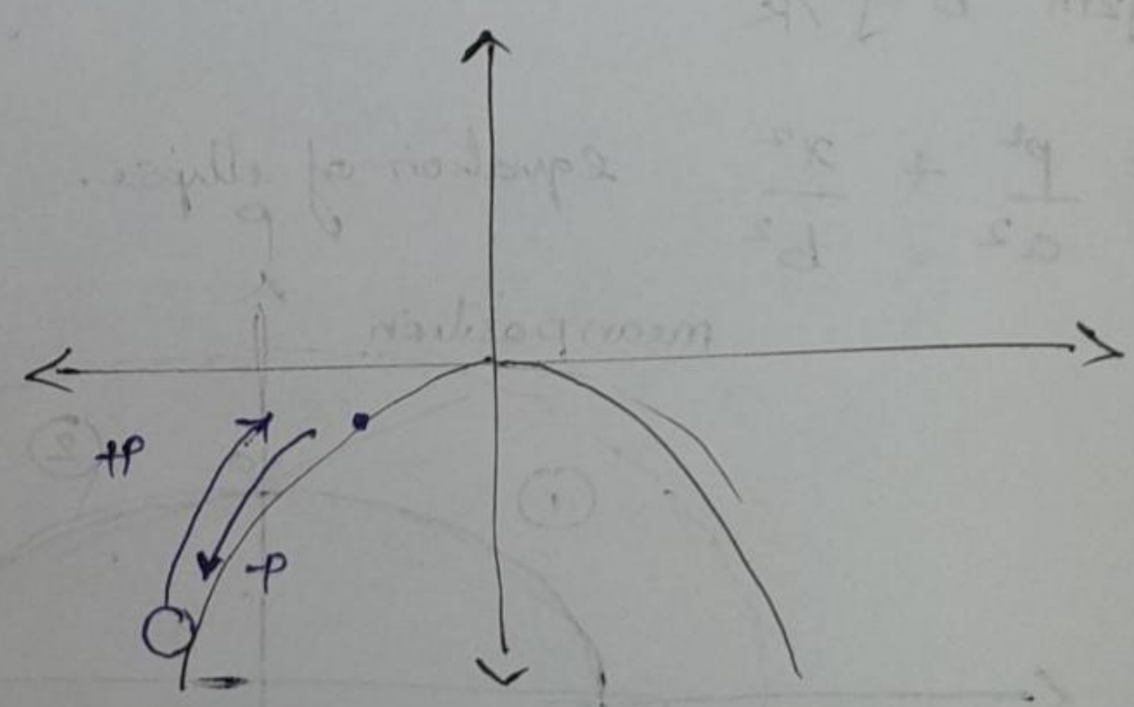
CASE 1



CASE 2



CASE 3



CASE 4

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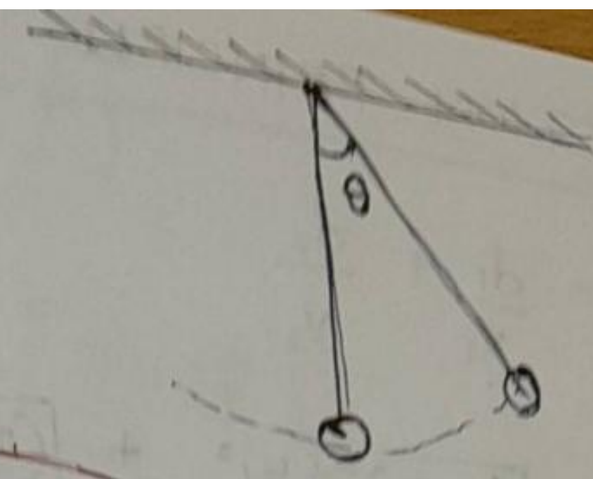
STANDARD

$E = \frac{1}{2} m l^2 \dot{\theta}^2 - mgl\cos\theta$

$\cos\theta = \frac{1-\theta^2}{2}$



when $\theta \simeq \theta$ comparable to $\frac{1}{2}mv^2 + mgl\theta_p^2$

$\dot{\theta} = \frac{d\theta}{dt}$

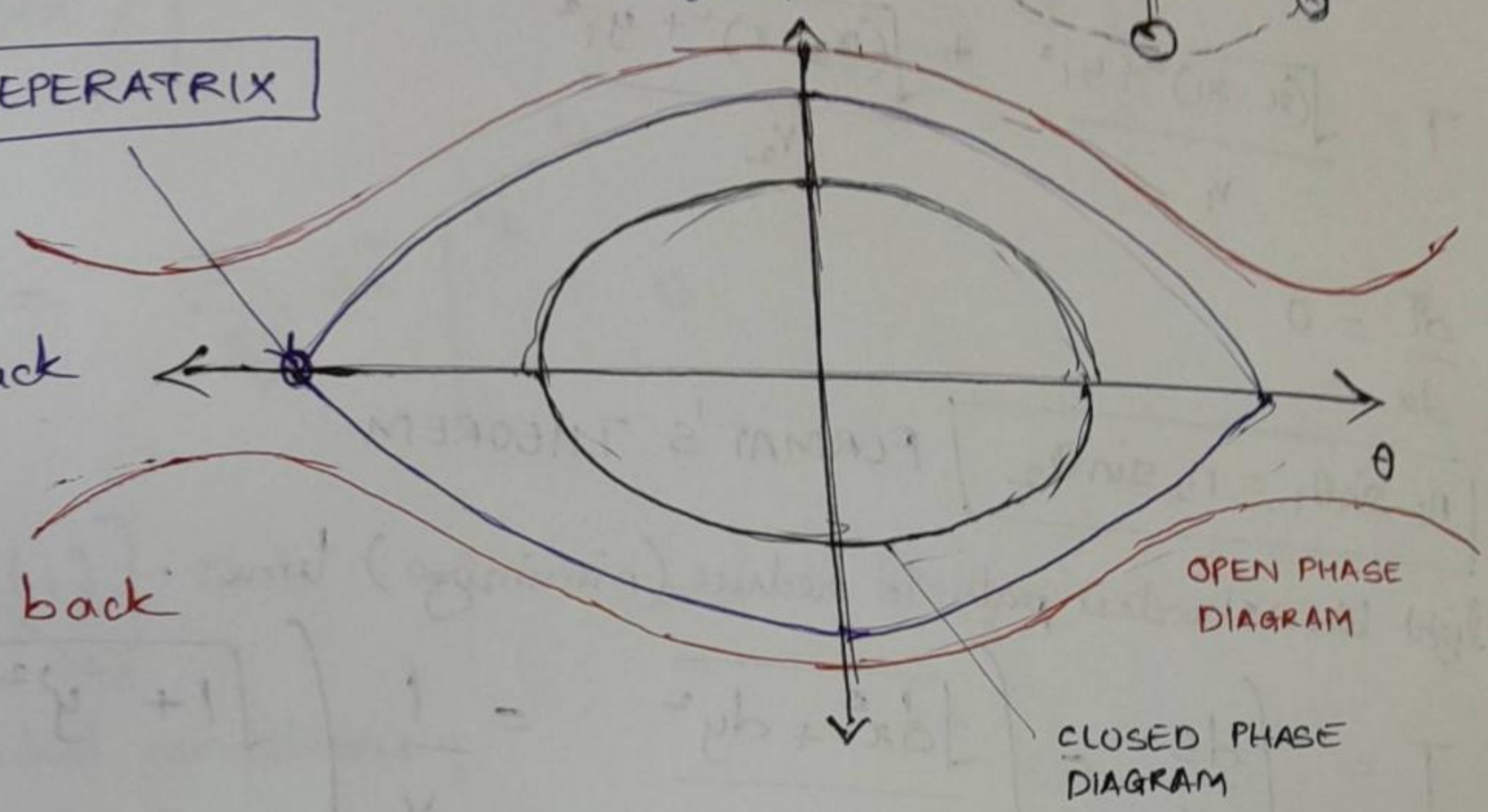
FOR $\theta \simeq \theta$ E_1

when drop from top and just stops and turns back

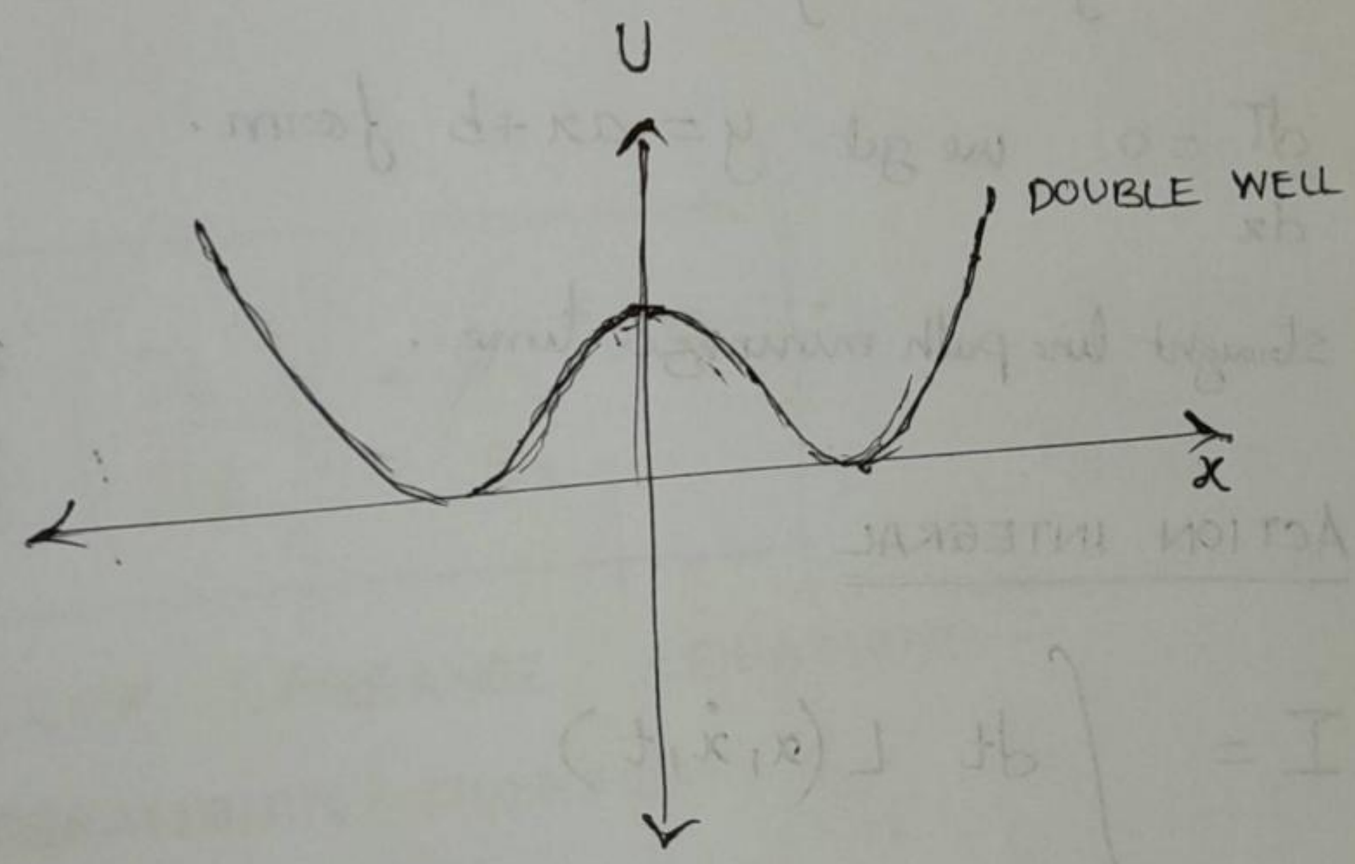
$E_2 > E_1$

high energy and pendulum rotates and does not turn back

$E_3 >> E_1$



DUFFING OSCILLATOR



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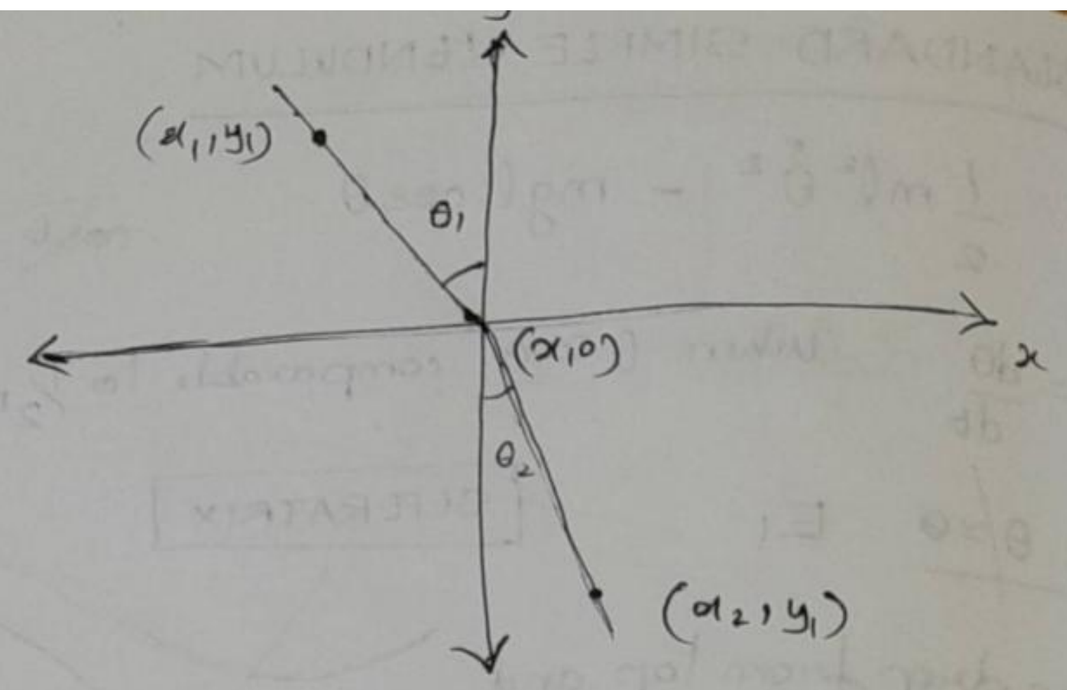
ACTION PRINCIPLE



$$T = \frac{d_1}{v_1} + \frac{d_2}{v_2}$$

$$T = \frac{\sqrt{(x-x_1)^2 + y_1^2}}{v_1} + \frac{\sqrt{(x_2-x)^2 + y_1^2}}{v_2}$$

$$\frac{dT}{dx} = 0$$

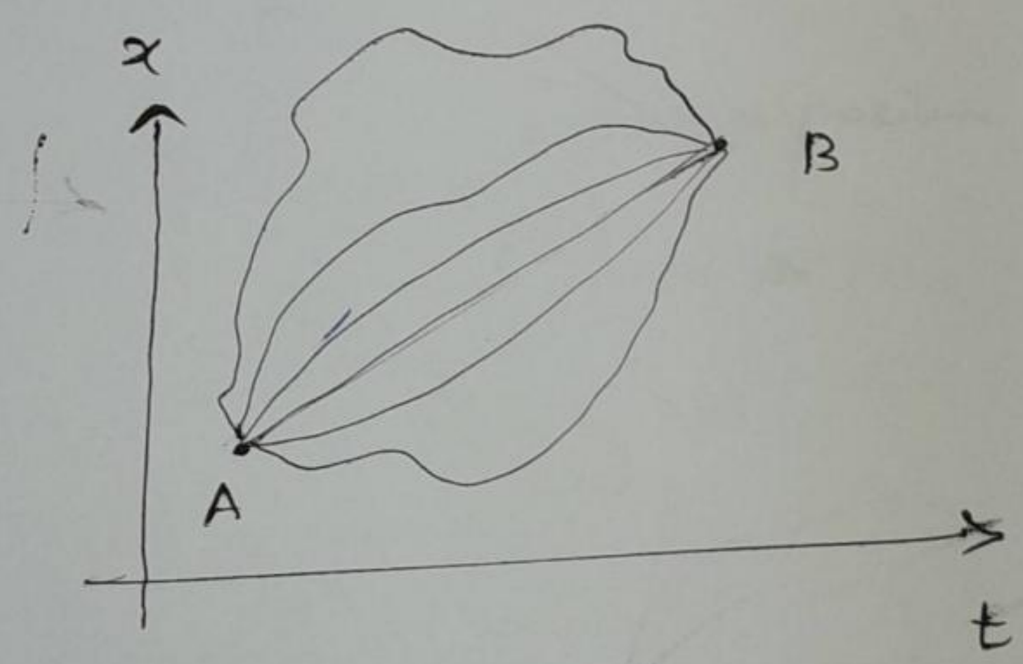
$$\boxed{n_1 \sin\theta_1 = n_2 \sin\theta_2}$$ FERMAT'S THEOREM

light takes shortest path to reduce (minimize) time.

$$T = \int \frac{ds}{v} = \int \frac{\sqrt{dx^2 + dy^2}}{v} = \frac{1}{v}\int \sqrt{1 + y'^2}\, dx \qquad y' = dy/dx$$

$\frac{dT}{dx} = 0$ we get $y = ax + b$ form.

straight line path minimizes time.



ACTION INTEGRAL

$$I = \int dt\; L(x, \dot{x}, t)$$

we want to extremize (minimize) this function.

δ is small change in ~~path~~ path.

$$\delta I = 0$$

$$\delta I = \int dt\; \delta L(x, \dot{x}, t) = \int dt \left[\frac{\partial L}{\partial x}\delta x + \frac{\partial L}{\partial \dot{x}}\delta \dot{x} + \cancel{\frac{\partial L}{\partial t}\delta t} \right]$$

$\to 0$ does not depend on time

$$\frac{\partial L}{\partial \dot{x}}\delta \dot{x} = \frac{\partial L}{\partial \dot{x}}\delta\left(\frac{dx}{dt}\right) = \underbrace{\frac{\partial L}{\partial \dot{x}}}_{A}\; \frac{d}{dt}\underbrace{(\delta x)}_{B} = A\frac{dB}{dt}$$

$$= \frac{d}{dt}(AB) - B\frac{dA}{dt} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\delta x\right) - \delta x\, \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right)$$

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$$\delta I = \int dt \left[\frac{\partial L}{\partial x}\delta x + \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\delta \dot{x}\right) - \delta \dot{x}\,\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right)\right]$$

$$= \int dt \left[\frac{\partial L}{\partial x} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right)\right]\delta x + \int_{t_1}^{t_2} dt\, \frac{d}{dt}\left[\frac{\partial L}{\partial \dot{x}}\delta x\right] \to 0$$

$$\int_{t_1}^{t_2} dt\, \frac{d}{dt}\left[\frac{\partial L}{\partial \dot{x}}\delta x\right] = \frac{\partial L}{\partial \dot{x}}\delta x \Big|_{t_1}^{t_2} = 0$$

$$\frac{\partial L}{\partial \dot{x}}\left[\delta x(t_2) - \delta x(t_1)\right] = 0$$

$\delta x(t_2) \to 0$, $\delta x(t_1) \to 0$ due to initial conditions.

and $\delta x|_{t_1} = 0$, $\delta x|_{t_2} = 0$

$$\delta I = dt\, \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = 0$$

If $\delta I = 0$ ★ $\boxed{\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = 0}$

$\boxed{\delta I = 0}$

EULER LAGRANGE EQUATIONS

LAGRANGIAN FUNCTION $= L$

Let $L = KE + PE = T - V$ $\quad T = \frac{1}{2}mv^2 \quad V = \frac{1}{2}mx^2$

$$\frac{d}{dt}\left[\frac{\partial (T-V)}{\partial \dot{x}}\right] - \frac{\partial (T-V)}{\partial x} = 0$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{x}}\right) - \frac{\partial V}{\partial x} = 0 \implies \frac{d}{dt}\left[m\dot{x}\right] + \frac{\partial V}{\partial x} = 0$$

assume $\frac{\partial T}{\partial x} = 0$, $\frac{\partial V}{\partial \dot{x}} = 0$

$m\ddot{x} = F = ma$ Newtons laws.

EULER LAGRANGE EQUATION in mechanics gives us Newton's laws

Newton's laws give us shortest path and not other paths.

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TUTORIAL 1

Q1 (a) $F_r \propto v \Rightarrow F_r = -kv = -mkv$

$F_r = ma \quad \cancel{m}\ddot{x} = -\cancel{m}kv$

$\frac{dv}{dt} = -kv \quad \frac{dv}{v} = -k\,dt \quad \ln v = -kt \quad v = e^{-kt} + \ln v_0$

to find integration constant $t = 0 \quad x = x_0, v = v_0$

$\ln\left(\frac{dx}{dt}\right) = -kt \quad \frac{dx}{dt} = e^{-kt} \quad x = -ke^{-kt} + 0$

$x = x_0 + \frac{-ke^{-kt}}{k}$

$\boxed{v = v_0 e^{-kt}} \quad \boxed{x = \frac{v_0}{k}(1 - e^{-kt})}$

(b) $\cancel{m}\ddot{x} = \cancel{m}g - \cancel{m}kv$

$\frac{dv}{dt} = g - vk \quad \frac{dv}{g-kv} = dt$

$\frac{dv}{g-kv} = dt \quad \frac{\ln(g-kv)}{\ln(-k)} = t$

$F = -mg - mkv \quad \frac{dv}{dt} = -g - kv \quad t_0 = 0$

$\cancel{m}\ddot{x} = -\cancel{m}g - \cancel{m}kv$

$\frac{dv}{g+kv} = -dt \quad \ln(g+kv) - \ln(g+kv_0) = -(t - t_0)$

$\ln\left(\frac{g+kv_0}{g+kv}\right)\frac{1}{k} = -t$

$\left(\frac{g+kv_0}{g+kv_0}\right) = e^{-kt} \quad g + kv = (g+kv_0)e^{-kt}$

$v = \frac{-g}{k} + \frac{(g+kv_0)}{k}e^{-kt}$

$\frac{dx}{dt} = \frac{-g}{k} + \left(\frac{g+kv_0}{k}\right)e^{-kt}$

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Q2 muzzle velocity $= V_0$ angle $= \theta$ $V_x = V_0 \cos\theta$

with air resistance $m\ddot{x} = -mkV_x$ $V_y = V_0 \sin\theta$

$m\ddot{y} = mg - mkV_y$

$V_x = \dot{x} = V_0 \cos\theta$ $V_y = \dot{y} = V_0 \sin\theta - gt$

$x = V_0 \cos\theta t$ $y = V_0 \sin\theta t - gt^2/2$

$\cancel{m}\ddot{x} = -\cancel{m}kV_x$

$\ddot{x} = -kV_0 \cos\theta$ $\frac{dv}{dt} = -kV_0 \cos\theta$ $\frac{dv}{v} = -k\cos\theta\, dt$

$\cancel{\frac{V_x}{V_0} = -k\cos\theta t}$ $\ln(V_x) = -k\cos\theta t$ $V_x = \cos\theta V_0 e^{-kt}$

$$x = \frac{V_0 \cos\theta}{k}\left(1 - e^{-kt}\right)$$

$$y = \frac{gt}{k} + \left(\frac{kV_0 \sin\theta + g}{k}\right)\left(1 - e^{-kt}\right)$$

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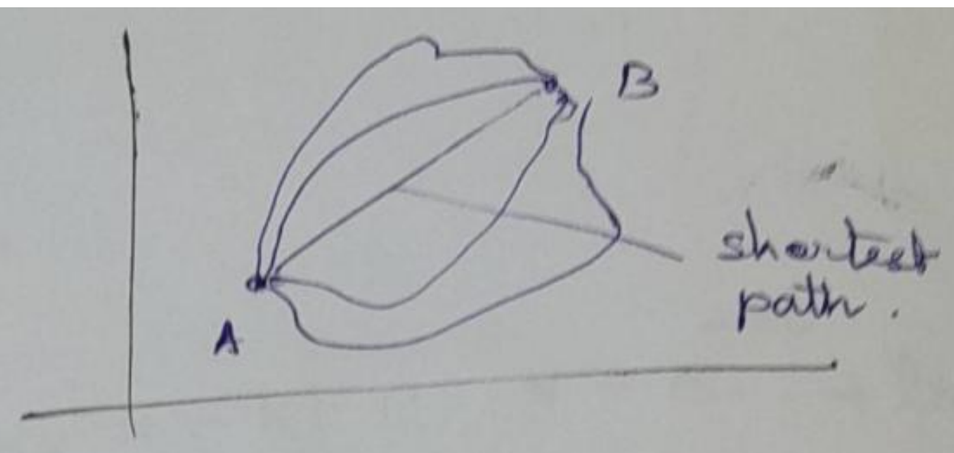
ACTION PRINCIPLE



$$I = \int_A^B dt \; L(x, \dot{x}, t)$$

$$I = \int dz \; f(y, y', z)$$

$$\delta I = 0$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = 0 \qquad \frac{d}{dz}\left(\frac{\partial f}{\partial y'}\right) - \frac{\partial f}{\partial y} = 0$$

Q1 (a) $T = \frac{1}{v}\int ds = \frac{1}{v}\int \sqrt{dx^2 + dy^2} = \frac{1}{v} \; dx \underbrace{\sqrt{1 + y'^2}}_{f(y, y', x)}$

$$f(y, y', x) = \sqrt{1 + y'^2}$$

$$\frac{\partial f}{\partial y} = 0, \; \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = \frac{\partial f}{\partial y'} = \text{constant}$$

$$\frac{\partial f}{\partial y'} = \frac{y'}{\sqrt{1 + y'^2}} = c \qquad y'^2 = c + c y'^2$$

$$y'^2 = c \quad y' = c \qquad \frac{dy}{dx} = c$$

$$dy = mx + c$$

∴ Shortest distance between 2 points in plane = line

$$\boxed{L = T - V}$$

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(i) FREE PARTICLE IN ONE DIMENSION

$V(x) = 0$

$L = \frac{1}{2}mv^2 = \frac{1}{2}m\dot{x}^2$

$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = 0$ but $\frac{\partial L}{\partial x} = 0$

$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = 0 \implies \frac{d}{dt}(m\dot{x}) = 0$ $\quad m\dot{x} = 0$

(ii) FREE PARTICLE IN TWO DIMENSION

$L = L(x, \dot{x}, y, \dot{y}, t) = T(\dot{x}, \dot{y})$

$L = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2$

$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = 0$ and $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{y}}\right) - \frac{\partial L}{\partial y} = 0$

from (i)

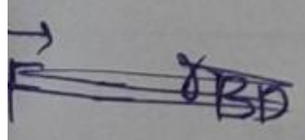
$m\dot{x} = 0$ $\quad m\dot{y} = 0$

(iii) $L = T - V = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$ (harmonic oscillator)

$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = 0 \implies \frac{d}{dt}(m\dot{x}) + kx = 0$

$\frac{d}{dt}\left(m\frac{dx}{dt}\right) + kx = 0$ $\quad (m\ddot{x}) = -kx$

$\ddot{x} = -\frac{k}{m}x$ $\quad \omega = \sqrt{\frac{k}{m}}$ $\quad \therefore \ddot{x} = -\omega^2 x$



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(IV) TWO DIMENSION (POLAR COORDINATES)

$x = r\cos\theta \quad y = r\sin\theta$

$x^2 + y^2 = r^2$

$L = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2$

$r(t), \theta(t)$

$L = L(r, \dot{r}, \theta, \dot{\theta})$

$L = \frac{m}{2}(\dot{r}^2 + r^2\dot{\theta}^2)$

$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) - \frac{\partial L}{\partial r} = 0$

$\frac{d}{dt}(m\dot{r}) - mr\dot{\theta}^2 = 0$

$\cancel{m}\ddot{r} - \cancel{m}r\dot{\theta}^2 = 0$

$\boxed{\ddot{r} = r\dot{\theta}^2}$

$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0$

$\frac{d}{dt}(mr^2\dot{\theta}) = 0$

$\frac{d}{dt}(mr^2\dot{\theta}) = 0$ angular momentum constant

$\boxed{mr^2\ddot{\theta} = 0}$

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CENTRAL FORCE

$$L = \frac{m}{2}(\dot{r}^2 + r^2\dot{\theta}^2) - V(r)$$

PRACTICE PROBLEM 2

Q3 (b) $L = \frac{m}{2}(\dot{x}^2 + \dot{y}^2) - mgy$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = 0$$

$$\frac{d}{dt}(m\dot{x}) = 0$$

$$\boxed{m\ddot{x} = 0}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{y}}\right) - \frac{\partial L}{\partial y} = 0$$

$$\frac{d}{dt}(m\dot{y}) + mg = 0$$

$$\cancel{m}\ddot{y} = -\cancel{m}g$$

$$\boxed{\ddot{y} = -g}$$

Q3 (d) $L = \frac{1}{2}ml^2\dot{\theta}^2 + mgl\cos\theta$

$$\cancel{\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{l}}\right) - \frac{\partial L}{\partial l} = 0}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0$$

$$\cancel{ml^2}\ \frac{\partial L}{\partial \theta} = \cancel{ml^2\theta}\ - mgl\sin\theta$$

$$\frac{\partial L}{\partial \dot{\theta}} = ml^2\dot{\theta}$$

$$\frac{d}{dt}(ml^2\dot{\theta}) + mgl\sin\theta = 0$$

$$l\ddot{\theta} = -g\sin\theta$$

TUTORIAL 1

1. $L = e^{\gamma t}\left(\frac{m\dot{q}^2}{2} - V(q)\right)$

$$m\ddot{x} + m\gamma\dot{x} + \frac{dV}{dx} = 0$$

damped

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16/03/2023

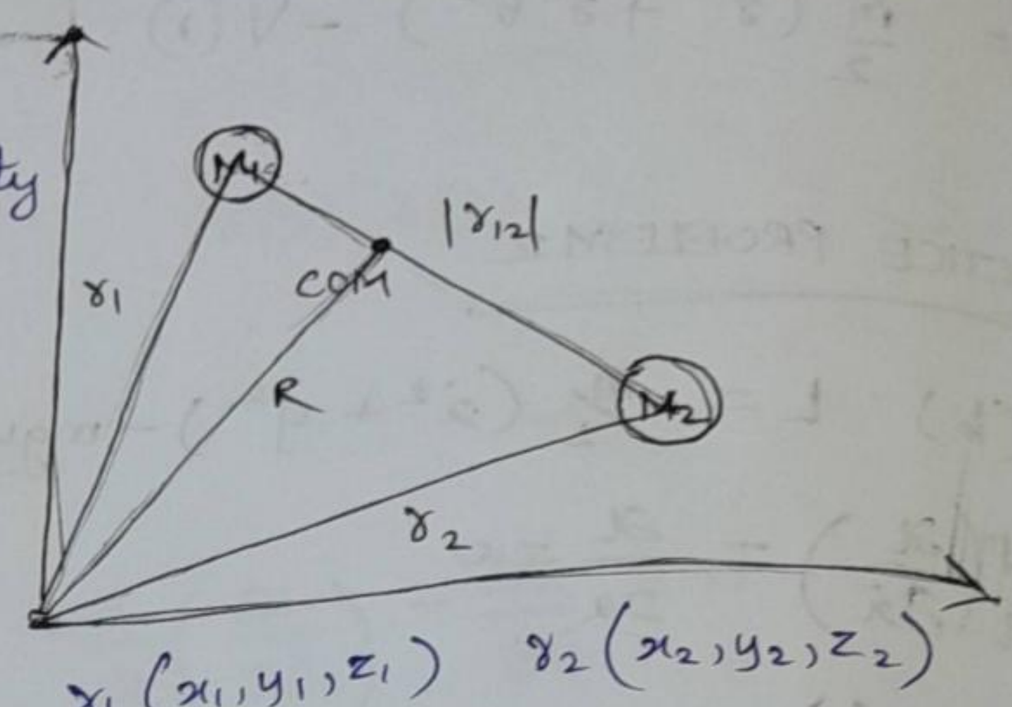
CENTRAL FORCES

$U(r) = -k/r$

Whenever central forces present

COM does not move with uniform velocity

6 degrees of freedom 3 of M_1 3 of M_2



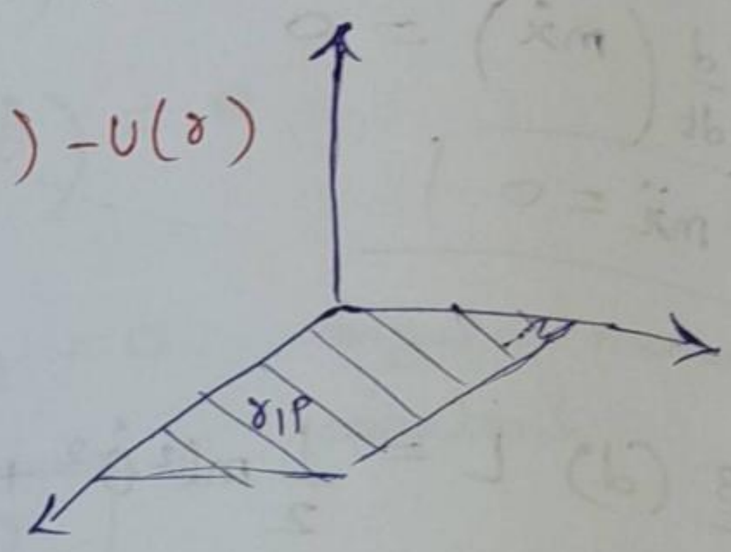
$mr^2\dot{\theta} = l$ = constant

$\vec{L} = \vec{r} \times \vec{p}$ if L in z axis r and p restricted to x-y plane

$$L = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) - U(r)$$

U(r) also has mass

$$L = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) - U(r)$$



mass of sun so high we take COM on sun itself

reduced mass $\mu = \dfrac{m_e m_s}{m_e + m_s} = \dfrac{m_e \cancel{m_s}}{\cancel{m_s}}$

$= m_e$

$$L = \frac{m_e}{2} (\dot{r}^2 + r^2 \dot{\theta}^2) - U(r)$$

by solving we get $r(t)$, $\theta(t)$ or $r(\theta)$

WHEN LAGRANGIAN IS (EXPLICITLY) INDEPENDENT OF A PARTICULAR COORDINATE THE CORRESPONDING CONJUGATE VARIABLE IS CONSERVED

$t \leftrightarrow E$

$x \leftrightarrow p$

$\theta \leftrightarrow L$

$$\frac{\partial L}{\partial t} = 0 \qquad H = \dot{q}\frac{\partial L}{\partial \dot{q}} - L = E$$

Hamiltonian

NOETHER'S THEOREM

if $\dfrac{\partial L}{\partial x} = 0$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = 0$$

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x} = p \qquad \frac{dp}{dt} = 0 \quad p \text{ constant.}$$

$$= \frac{1}{2} m\dot{x}^2 - U(x) \qquad H = \frac{1}{2} m\dot{x}^2 + U(x)$$

LEGENDRE TRANSFORMATION

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conserved quantity $\mu r^2 \dot{\theta} = l$

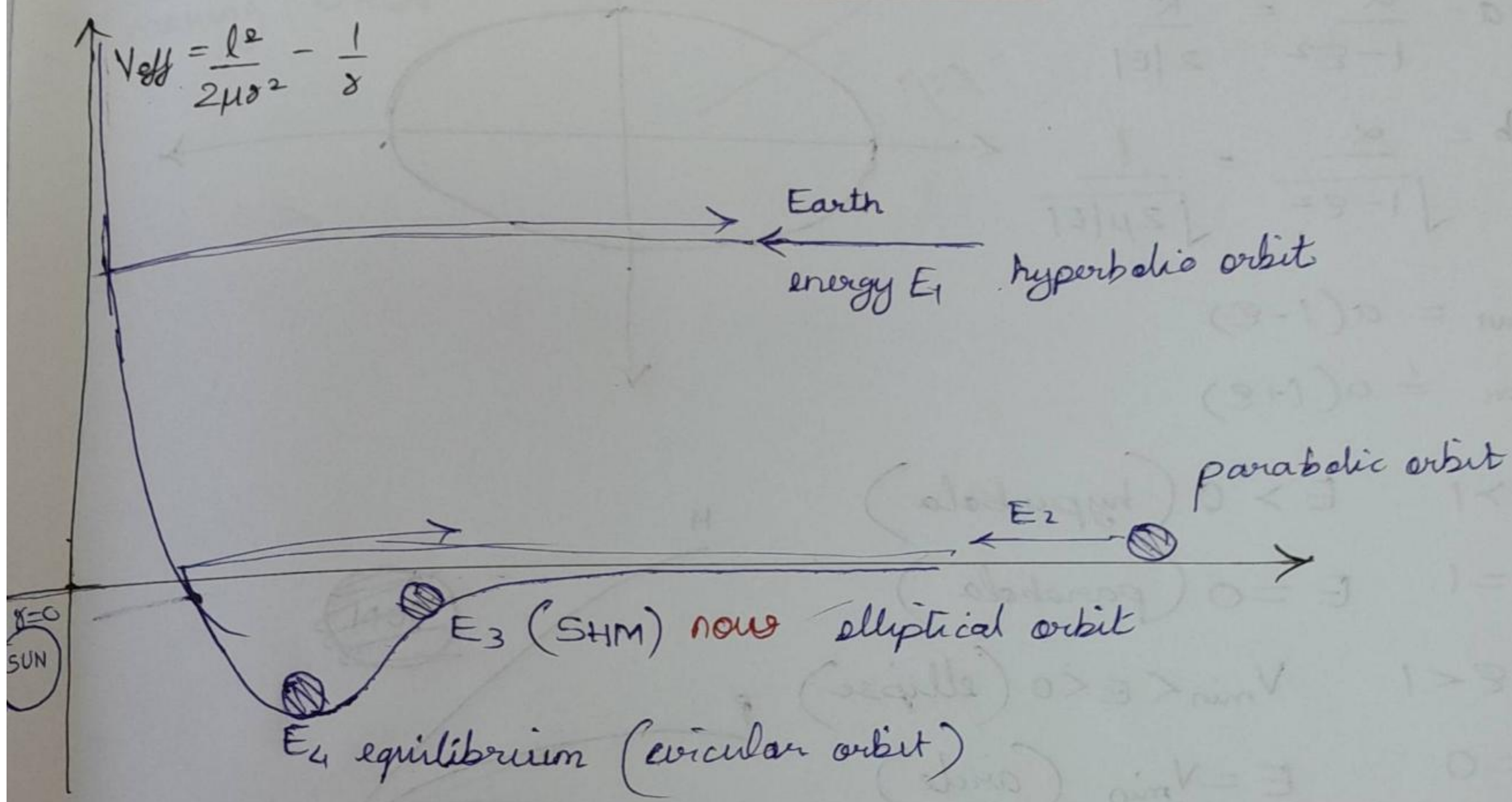
$$E = \frac{1}{2}\mu(\dot{r}^2 + r^2\dot{\theta}^2) - k/r$$

$$E = \frac{1}{2}\mu\dot{r}^2 + \frac{1}{2}\mu r^2\dot{\theta}^2 - \frac{k}{r} = \frac{1}{2}\mu\dot{r}^2 + \frac{1}{2}\cancel{\mu r^2}\frac{l^2}{\mu^{\cancel{2}} r^{\cancel{4}2}} - \frac{k}{r}$$

$$= \frac{1}{2}\mu\dot{r}^2 + \frac{l^2}{2\mu r^2} - \frac{k}{r}$$

$$\boxed{E = \frac{\mu\dot{r}^2}{2} + V_{eff}(r)}$$

effective one dimension problem



$$\mu r^2 \dot{\theta} = l$$
$$\mu r^2 \frac{d\theta}{dt} = l \implies \int d\theta = 0 \int \frac{dt}{\mu r^2(t)} = \frac{l}{\mu r^2} dt$$

$$\theta(t) = at + b$$

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$$\frac{\alpha}{r} = 1 + \varepsilon \cos\theta \qquad \alpha = \frac{l^2}{\mu k}$$

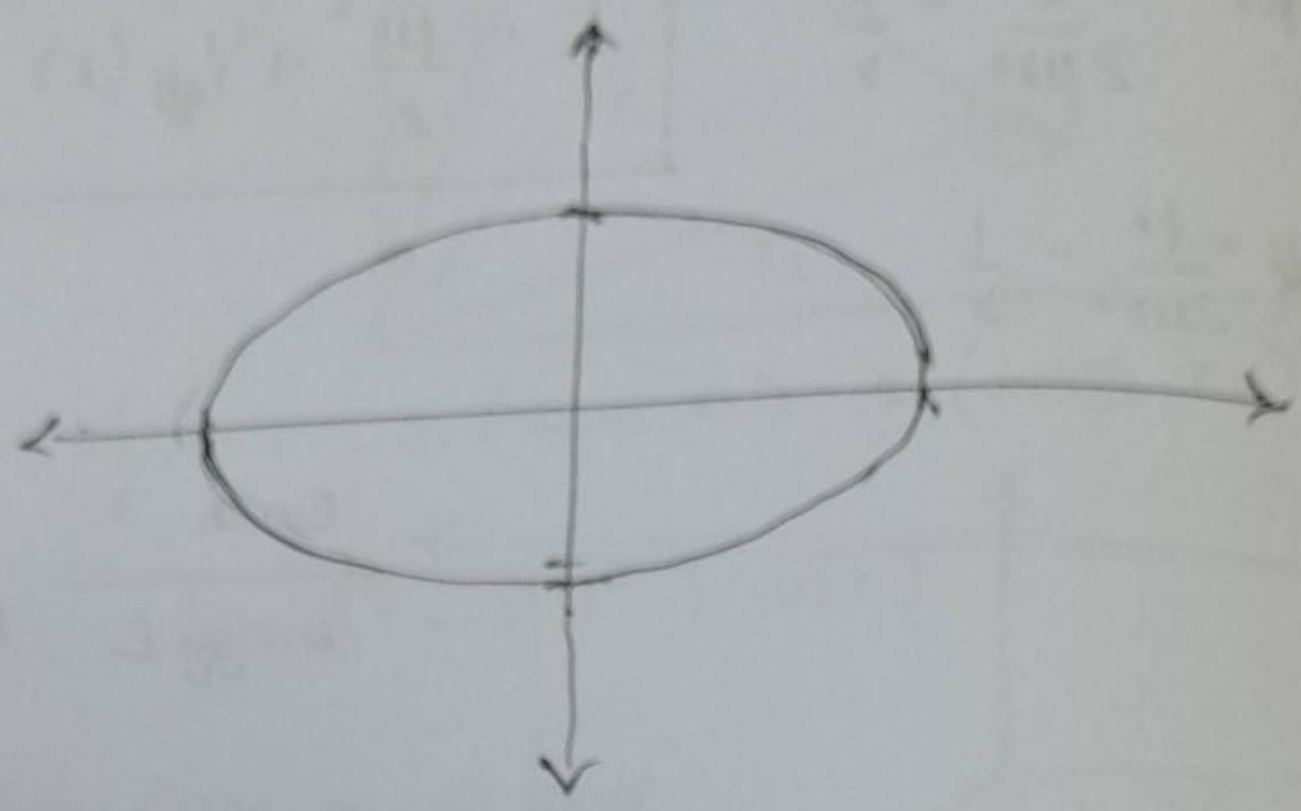
Conic section

$$\varepsilon = \sqrt{1 + \frac{2El^2}{\mu k^2}}$$

<u>ELLIPSE</u>

$$a = \frac{\alpha}{1-\varepsilon^2} = \frac{k}{2|E|}$$

$$b = \frac{\alpha}{\sqrt{1-\varepsilon^2}} = \frac{l}{\sqrt{2\mu|E|}}$$



$$r_{min} = a(1-\varepsilon)$$

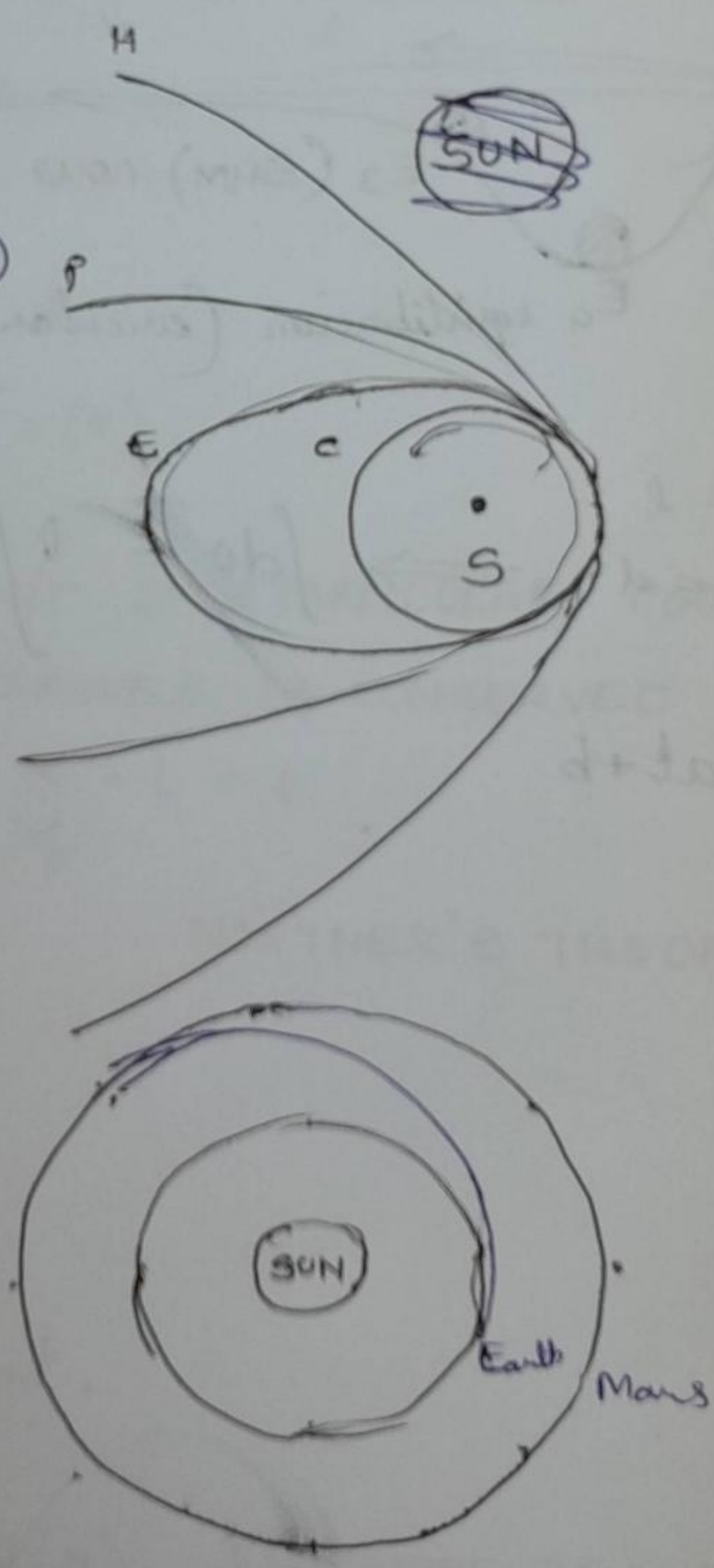
$$r_{max} = a(1+\varepsilon)$$

$\varepsilon > 1 \qquad E > 0$ (hyperbola)

$\varepsilon = 1 \qquad E = 0$ (parabola)

$0 < \varepsilon < 1 \qquad V_{min} < E < 0$ (ellipse)

$E = 0 \qquad E = V_{min}$ (circle)



$$E = -k/2a$$

$$E = -k/2r_1 = \frac{1}{2} m v_1^2 - \frac{k}{r_1}$$

$$v_1 = \sqrt{\frac{k}{m r_1}}$$

Hohmann transfer

$$E_t = \frac{-k}{(r_1 + r_2)} = \frac{1}{2} m v_{t_1}^2 - k/r_1$$

$$v_{t_1} = \sqrt{\frac{2k}{m r_1}\left(\frac{r_2}{r_1 + r_2}\right)}$$

$$\Delta v_1 = v_{t_1} - v_1$$

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SPECIAL RELATIVITY

GALILEAN TRANSFORMATION

$x' = x - vt$

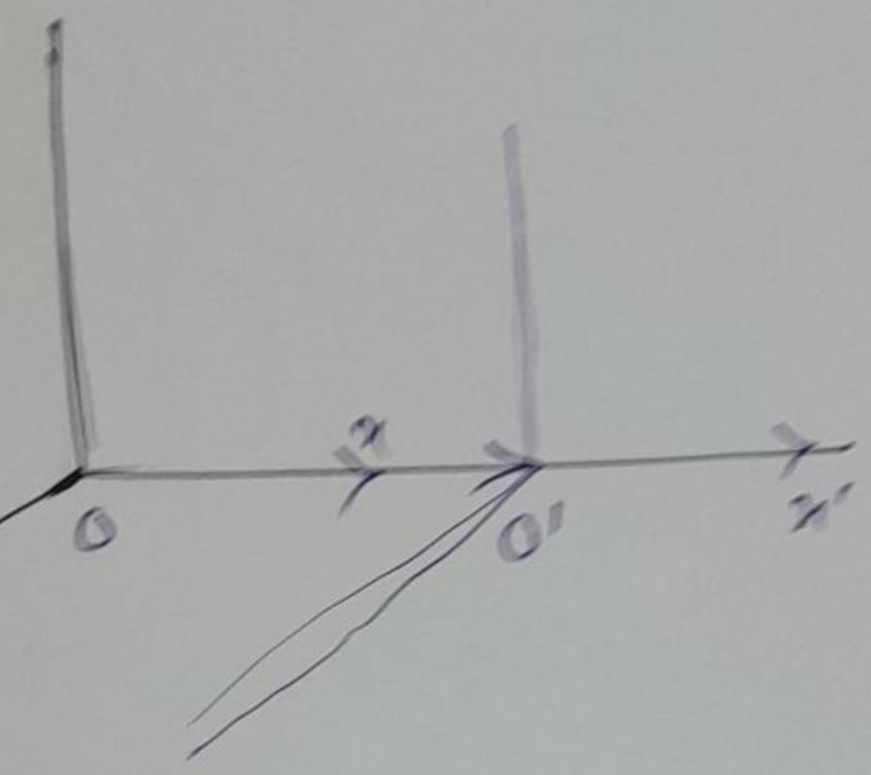
$y' = y$

$z' = z$

$\boxed{t' = t}$ → simultaneity problem

of meaning

O rest



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